Threshold-Set Diagnostic Study

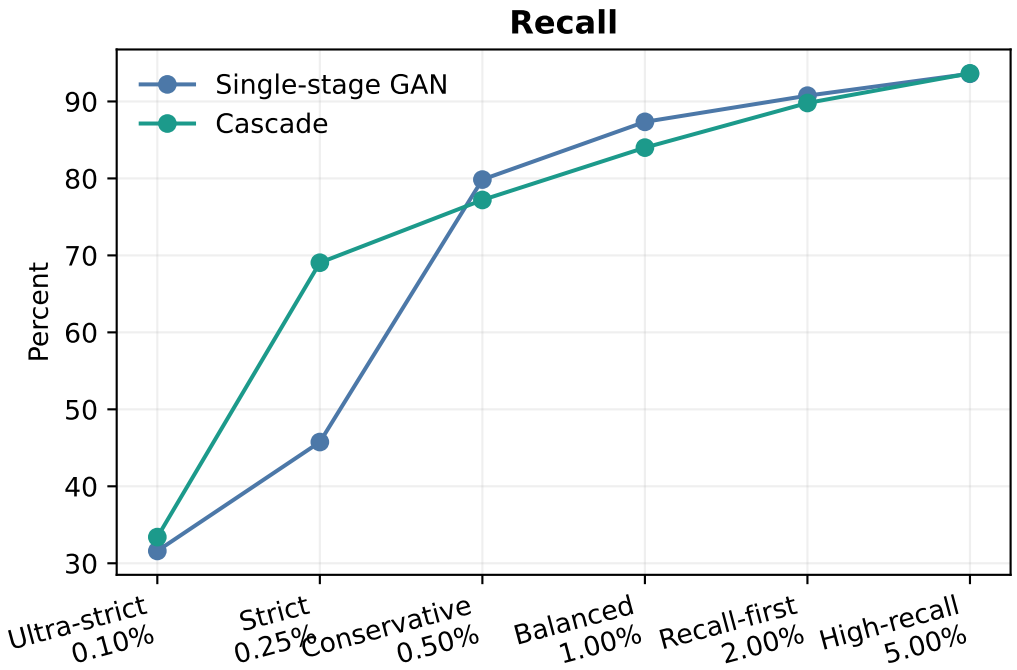
Frozen HYB-MS +10% GAN and two-stage cascade • 20,000 identity-disjoint URLs

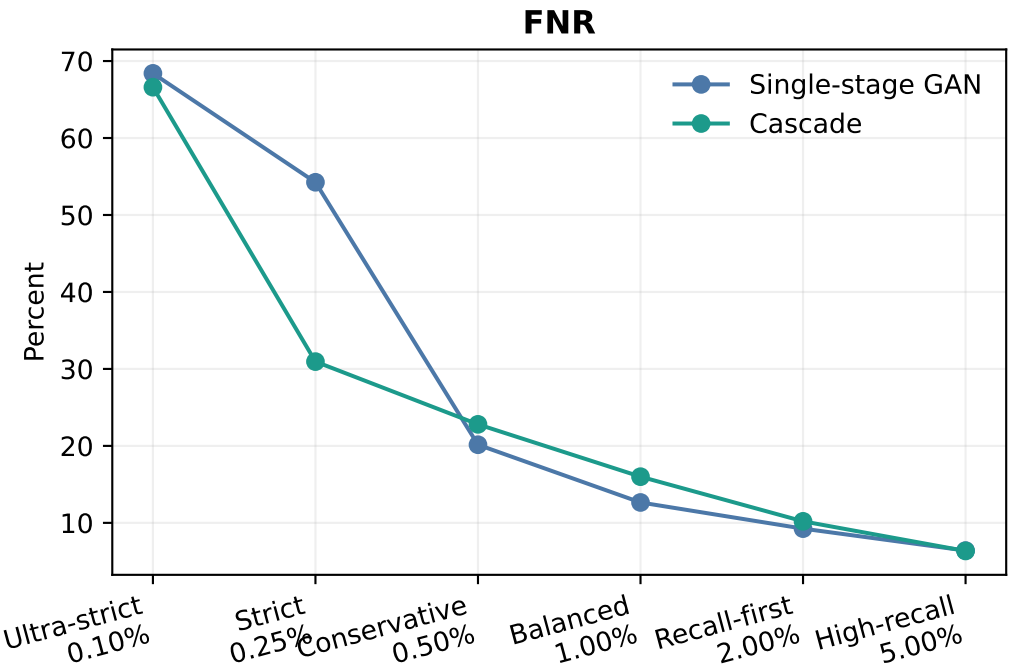
Set	Cal. cap	System	Threshold	Precision	Recall	F1	FPR	FNR
Ultra-strict	0.10%	Single	0.999973	99.37%	31.60%	47.95%	0.02%	68.40%
		Cascade	0.999980	99.40%	33.40%	50.00%	0.02%	66.60%
Strict	0.25%	Single	0.999849	98.60%	45.75%	62.50%	0.07%	54.25%
		Cascade	0.988409	98.71%	69.05%	81.26%	0.10%	30.95%
Conservative	0.50%	Single	0.995319	94.83%	79.85%	86.70%	0.48%	20.15%
		Cascade	0.966013	96.98%	77.20%	85.97%	0.27%	22.80%
Balanced	1.00%	Single	0.970766	89.77%	87.35%	88.55%	1.11%	12.65%
		Cascade	0.888736	90.96%	84.00%	87.34%	0.93%	16.00%
Recall-first	2.00%	Single	0.885636	81.57%	90.75%	85.92%	2.28%	9.25%
		Cascade	0.581254	82.99%	89.80%	86.26%	2.04%	10.20%
High-recall	5.00%	Single	0.670915	64.84%	93.60%	76.61%	5.64%	6.40%
		Cascade	0.062300	65.56%	93.65%	77.13%	5.47%	6.35%

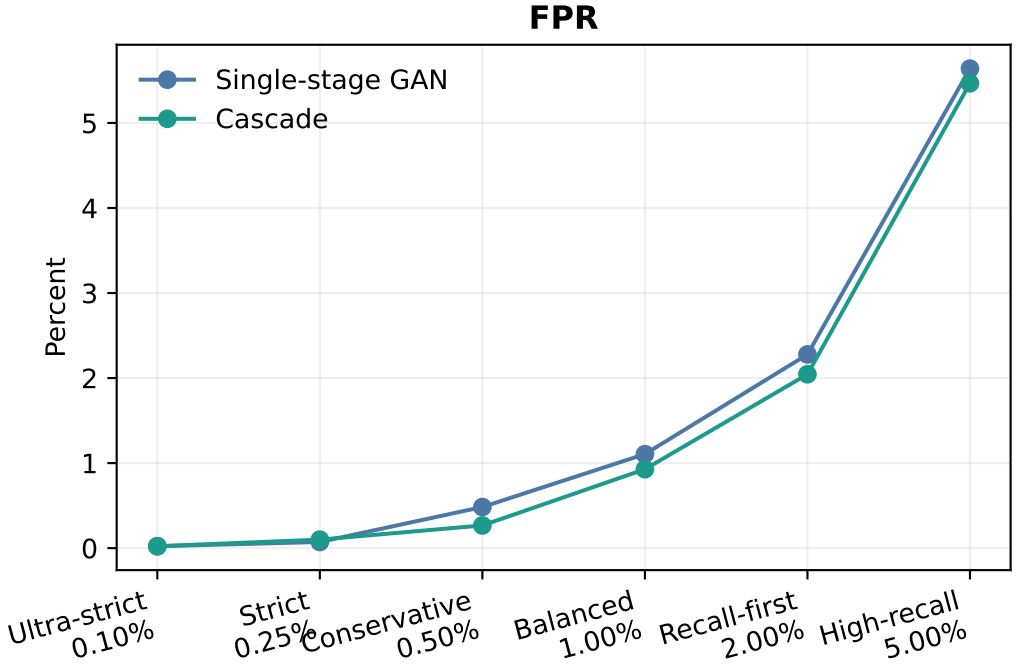
The lowest observed FNR is 6.35%: High-recall cascade, with observed FPR 5.47%. No tested threshold eliminates false negatives.

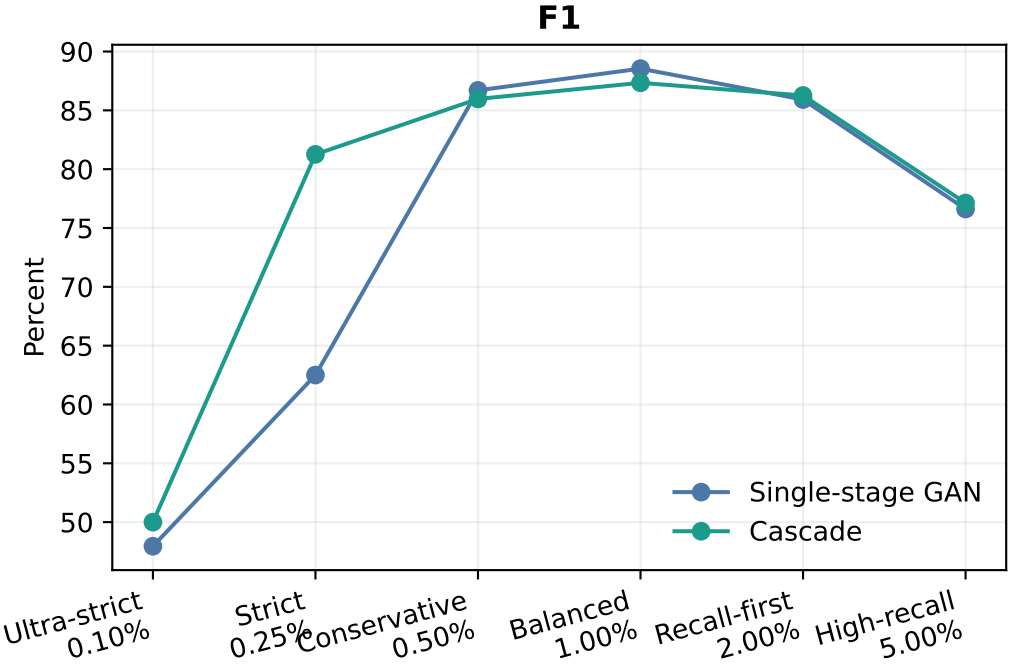
Thresholds were derived only from calibration data. Because this 20K cohort was previously evaluated, this sweep is diagnostic and cannot be presented as a new untouched confirmation.

What changes when the trust threshold changes?









Single-Stage Score Bands

Band	Lower score	Upper score	URLs	Legitimate	Phishing	Observed phish %
Low	−∞	0.670915	17,113	16,985	128	0.75%
Uncertain	0.670915	0.970766	941	816	125	13.28%
Likely phishing	0.970766	0.999849	1,018	186	832	81.73%
High risk	0.999849	0.999973	292	9	283	96.92%
Extreme risk	0.999973	+∞	636	4	632	99.37%

Interpretation

A phishing URL in the low band is not a threshold problem: even the high-recall threshold gives it a low score. Those cases point to missing representation or distribution coverage. Phishing concentrated just below a threshold can be recovered by lowering that threshold, at the cost shown by the legitimate counts entering higher bands.

A score is model confidence under this training/calibration setup, not a guaranteed probability of maliciousness. The observe percentage in each band depends on this deliberately 10%-phishing evaluation prevalence.

Marginal Security Cost

Set	Single: extra TP	Single: extra FP	TP per extra FP	Cascade: extra TP	Cascade: extra FP	TP per extra FP
Ultra-strict			—			—
Strict	283	9	31.44	713	14	50.93
Conservative	682	74	9.22	163	30	5.43
Balanced	150	112	1.34	136	119	1.14
Recall-first	68	211	0.32	116	201	0.58
High-recall	57	605	0.09	77	616	0.12

Operational reading

Use the Strict cascade when approximately 0.1% observed FPR is required. Use the Balanced or Recall-first single-stage setting when reducing missed phishing is more important and the extension can tolerate additional warnings. Treat High-recall as a diagnostic boundary, not an automatic deployment default.

Projected precision at 0.1%, 1%, 5%, and 10% phishing prevalence is stored in the JSON and CSV. Precision observed on this 10% cohort should not be assumed for ordinary browser traffic.